Worksheet 4.4 — Applications of Exponential & Log Models

Grade 12 Advanced Functions (MHF4U) · Unit 4: Exponential and Logarithmic Functions

The Big Ideas

Model with $A = A_0\, b^{t/p}$ (growth $b > 1$, decay $0 < b < 1$); use logs for doubling time and half-life; log scales (pH, dB, Richter) step by factors of ten. State the model, name what each letter means, and write the **whole model** in every step.

- $A = A_0\, b^{t/p}$: A_0 is the start, b the factor per period (2 for doubling, $\frac{1}{2}$ for half-life), p the length of a period.
- Finding a time: isolate the power, then match bases or take logs. Doubling time $= \dfrac{\log 2}{\log(\text{growth factor})}$.
- $\text{pH} = -\log[\text{H}^+]$; each Richter step is $\times 10$, so a difference of d steps is 10^d.

Core Concepts

Growth and decay

Exponential models $A = A_0 b^t$ describe populations, radioactive decay, and cooling; logarithms solve them for the time.

Compound interest

Investments grow by $A = P(1+i)^n$, and finding the number of periods n requires a logarithm.

Logarithmic scales

Quantities spanning huge ranges sound in decibels, earthquakes on the Richter scale, acidity as pH use **logarithmic** scales.

Writing solutions

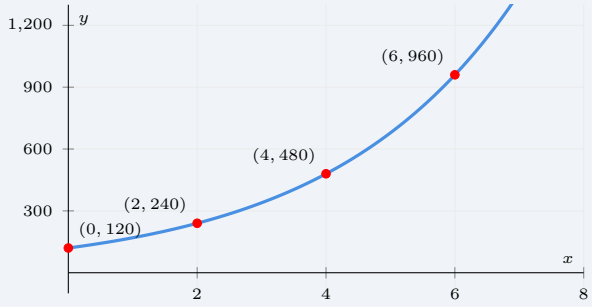
State the model and what each letter means, then write the **whole model** in every step (for example $P = 100 \cdot 2^{10/5}$). If the target is a whole number of periods, match the bases; otherwise take logs, and always check by substituting back.

Worked Examples

Example 1 — Growth

$P = 120 \cdot 2^{t/2}$. Find P at $t = 6$.

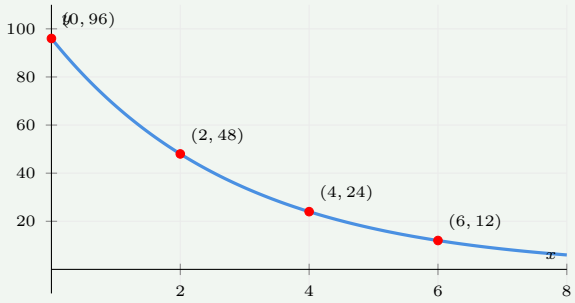
Solution. **Step 1 (read the model):** In $P = 120 \cdot 2^{t/2}$: $P_0 = 120$, the factor is 2 (doubling) and the period is $p = 2$.
Step 2 (substitute): Put $t = 6$ into $P = 120 \cdot 2^{t/2}$: $P = 120 \cdot 2^{6/2} = 120 \cdot 2^3$.
Step 3 (evaluate): $120 \cdot 8 = 960$.
Step 4 (check): Three periods pass, so the amount doubles three times: $120 \to 240 \to 480 \to 960$ ✓.
Answer: $P = 960$.



Example 2 — Half-life

$A = 96\left(\frac{1}{2}\right)^{t/2}$. Find A at $t = 6$.

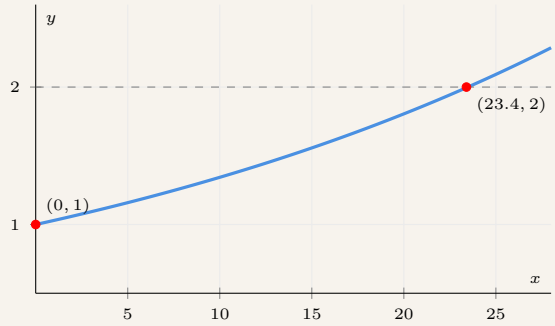
Solution. **Step 1 (read the model):** In $A = 96\left(\frac{1}{2}\right)^{t/2}$: $A_0 = 96$, the factor is $\frac{1}{2}$ (halving) and the half-life is $p = 2$.
Step 2 (substitute): Put $t = 6$ into $A = 96\left(\frac{1}{2}\right)^{t/2}$: $A = 96\left(\frac{1}{2}\right)^{6/2} = 96\left(\frac{1}{2}\right)^3$.
Step 3 (evaluate): $96 \cdot \frac{1}{8} = 12$.
Step 4 (check): Three half-lives: $96 \to 48 \to 24 \to 12$ ✓.
Answer: $A = 12$ mg.



Example 3 — Doubling time

How long to double at 3% /yr?

Solution. **Step 1 (model):** The yearly growth factor is 1.03, so $A = A_0(1.03)^t$.
Step 2 (set up the doubling): Doubling means $A = 2A_0$: $2A_0 = A_0(1.03)^t$, so $2 = 1.03^t$.
Step 3 (take logs): Take the logarithm of both sides of $2 = 1.03^t$: $\log 2 = t \log 1.03$.
Step 4 (solve): $t = \dfrac{\log 2}{\log 1.03} \approx \dfrac{0.3010}{0.01284} \approx 23.4$.
Step 5 (check): $1.03^{23.4} \approx 2.00$ ✓.
Answer: about 23.4 years.



Example 4 — pH scale

Find the pH if $[H^+] = 10^{-6}$.

Solution. **Step 1 (substitute):** $\text{pH} = -\log[H^+]$ becomes $\text{pH} = -\log(10^{-6})$.
Step 2 (evaluate): $\log(10^{-6}) = -6$, so $\text{pH} = -(-6) = 6$.
Answer: $\text{pH} = 6$.

Example 5 — Richter scale

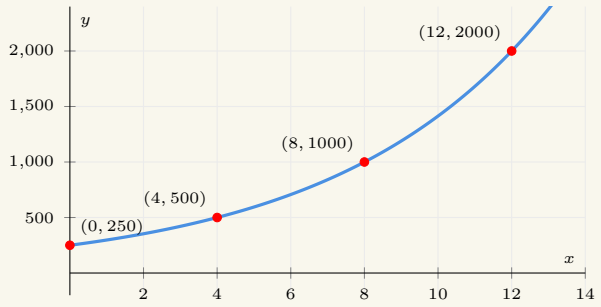
How much stronger is magnitude 8 than magnitude 7?

Solution. **Step 1 (the scale):** Each unit of magnitude multiplies the amplitude by 10, so the ratio is 10^d where d is the difference.
Step 2 (difference): $d = 8 - 7 = 1$.
Step 3 (ratio): $10^d = 10^1 = 10$.
Answer: $10\times$ the amplitude.

Example 6 — Growth

$P = 250 \cdot 2^{t/4}$. Find P at $t = 12$.

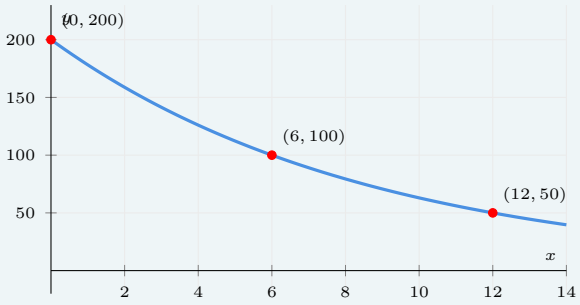
Solution. **Step 1 (read the model):** In $P = 250 \cdot 2^{t/4}$: $P_0 = 250$, doubling, period $p = 4$.
Step 2 (substitute): Put $t = 12$ into $P = 250 \cdot 2^{t/4}$: $P = 250 \cdot 2^{12/4} = 250 \cdot 2^3$.
Step 3 (evaluate): $250 \cdot 8 = 2000$.
Step 4 (check): Three doublings: $250 \to 500 \to 1000 \to 2000$ ✓.
Answer: $P = 2000$.



Example 7 — Half-life

$A = 200\left(\frac{1}{2}\right)^{t/6}$. Find A at $t = 12$.

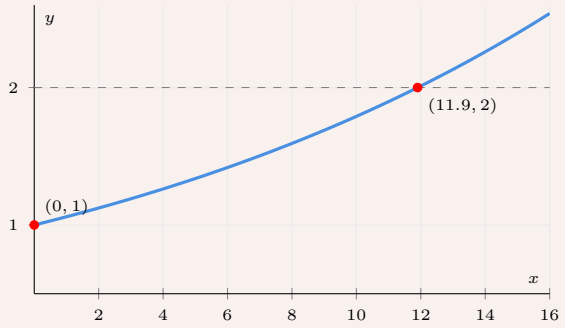
Solution. **Step 1 (read the model):** In $A = 200\left(\frac{1}{2}\right)^{t/6}$: $A_0 = 200$, halving, half-life $p = 6$.
Step 2 (substitute): Put $t = 12$ into $A = 200\left(\frac{1}{2}\right)^{t/6}$: $A = 200\left(\frac{1}{2}\right)^{12/6} = 200\left(\frac{1}{2}\right)^2$.
Step 3 (evaluate): $200 \cdot \frac{1}{4} = 50$.
Step 4 (check): Two half-lives: $200 \to 100 \to 50$ ✓.
Answer: $A = 50$.



Example 8 — Doubling time

How long to double at 6% /yr?

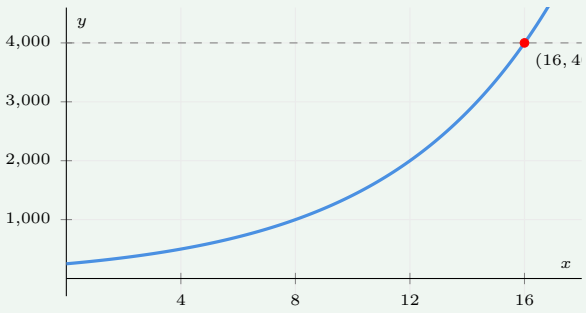
Solution. **Step 1 (model):** The yearly growth factor is 1.06, so $A = A_0(1.06)^t$.
Step 2 (set up the doubling): $2A_0 = A_0(1.06)^t$, so $2 = 1.06^t$.
Step 3 (take logs and solve): Take logs of both sides of $2 = 1.06^t$: $\log 2 = t \log 1.06$, so $t = \dfrac{\log 2}{\log 1.06} \approx \dfrac{0.3010}{0.02531} \approx 11.9$.
Step 4 (check): $1.06^{11.9} \approx 2.00$ ✓.
Answer: about 11.9 years.



Example 9 — Solve for time

$P = 250 \cdot 2^{t/4}$. When does P reach 4000?

Solution. **Step 1 (set up):** Set $P = 4000$ in $P = 250 \cdot 2^{t/4}$: $250 \cdot 2^{t/4} = 4000$.
Step 2 (isolate the power): Divide both sides of $250 \cdot 2^{t/4} = 4000$ by 250: $2^{t/4} = 16$.
Step 3 (common base): $16 = 2^4$, so $2^{t/4} = 2^4$.
Step 4 (equate the exponents): $\dfrac{t}{4} = 4 \Rightarrow t = 16$.
Step 5 (check): Four doublings: $250 \to 500 \to 1000 \to 2000 \to 4000$ ✓.
Answer: $P = 4000$ at $t = 16$ (no logs needed).



Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

Question 1

$P = 300 \cdot 2^{t/6}$. Find P at $t = 12$.

Question 2

$A = 240(\frac{1}{2})^{t/4}$. Find A at $t = 8$.

Question 3

How long to double at 8% /yr (2 d.p.)?

Question 4

pH if $[H^+] = 10^{-3}$?

Question 5

How much stronger is magnitude 6 than magnitude 3?

Question 6

$P = 100 \cdot 2^{t/10}$. Find P at $t = 20$.

Question 7

$A = 64(\frac{1}{2})^{t/2}$. Find A at $t = 6$.

Question 8

How long to triple at 5% /yr (2 d.p.)?

Question 9

pH if $[H^+] = 10^{-9}$?

Question 10

A quake is 10^4 times stronger than another. What is the magnitude difference?

Question 11

$P = 50 \cdot 3^{t/5}$. Find P at $t = 5$.

Question 12

How long for 5% /yr growth to reach 1.5 times the start (2 d.p.)?

Question 13 — Challenge

Bacteria double every 3 h from 500. Write the model and find the count at $t = 9$ h.

Question Recap & Answer Key

Question Recap

1. $P = 300 \cdot 2^{t/6}$. Find P at $t = 12$.
2. $A = 240(\frac{1}{2})^{t/4}$. Find A at $t = 8$.
3. How long to double at 8% /yr (2 d.p.)?
4. pH if $[\mathrm{H}^+] = 10^{-3}$?
5. How much stronger is magnitude 6 than magnitude 3?
6. $P = 100 \cdot 2^{t/10}$. Find P at $t = 20$.
7. $A = 64(\frac{1}{2})^{t/2}$. Find A at $t = 6$.
8. How long to triple at 5% /yr (2 d.p.)?
9. pH if $[\mathrm{H}^+] = 10^{-9}$?
10. A quake is 10^4 times stronger than another. What is the magnitude difference?
11. $P = 50 \cdot 3^{t/5}$. Find P at $t = 5$.
12. How long for 5% /yr growth to reach 1.5 times the start (2 d.p.)?
13. Bacteria double every 3 h from 500. Write the model and find the count at $t = 9$ h.

✓ Answer Key

1. $P = 300 \cdot 2^{12/6} = 300 \cdot 2^2 = 1200$
2. $A = 240\left(\frac{1}{2}\right)^{8/4} = 240\left(\frac{1}{2}\right)^2 = 60$
3. Solve $2 = 1.08^t$: $t = \dfrac{\log 2}{\log 1.08} \approx 9.01$ yr
4. $\mathrm{pH} = -\log(10^{-3}) = 3$
5. $10^{6-3} = 10^3 = 1000\times$
6. $P = 100 \cdot 2^{20/10} = 100 \cdot 2^2 = 400$
7. $A = 64\left(\frac{1}{2}\right)^{6/2} = 64\left(\frac{1}{2}\right)^3 = 8$
8. Solve $3 = 1.05^t$: $t = \dfrac{\log 3}{\log 1.05} \approx 22.52$ yr
9. $\mathrm{pH} = -\log(10^{-9}) = 9$
10. $10^d = 10^4 \Rightarrow d = 4$
11. $P = 50 \cdot 3^{5/5} = 50 \cdot 3 = 150$
12. Solve $1.5 = 1.05^t$: $t = \dfrac{\log 1.5}{\log 1.05} \approx 8.31$ yr
13. Model $P = 500 \cdot 2^{t/3}$; $P(9) = 500 \cdot 2^{9/3} = 500 \cdot 2^3 = 4000$